CSE7302- **University Project**

Review-5

Automated Student Attendance Monitoring And Analytics System For College

Batch Number: CSE_08

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Problem Statement

Organization: Government of Punjab

Category (Hardware / Software / Both) :Software

Problem Description :

- Most colleges still use manual or semi-automated attendance methods.
- These methods are time-consuming and increase teachers' workload.
- Attendance records are prone to human errors and data loss.
- Proxy attendance is possible in traditional systems.
- There is no centralized system to manage attendance data.
- Students do not get real-time access to their attendance status.
- Existing systems lack analytics and reporting features.
- It is difficult to identify students with low attendance early.
- Administrative decision-making is not data-driven.
- Hence, an automated and analytical attendance system is needed.

Objectives

- To automate the process of student attendance recording.
- To reduce manual work and human errors.
- To prevent proxy attendance and improve accuracy.
- To maintain a centralized and secure attendance database.
- To provide real-time access to attendance for students and faculty.
- To generate automated reports and summaries.
- To analyze attendance patterns and trends.
- To identify students with low attendance at an early stage.
- To improve transparency in attendance management.
- To support better academic and administrative decision-making.

Source code

INSTALL REQUIREMENTS

pip install fastapi uvicorn face_recognition opencv-python numpy sqlalchemy DATABASE (models.py)

from sqlalchemy import Column, String, DateTime from sqlalchemy.ext.declarative import declarative_base import uuid from datetime import datetime Base = declarative_base() class Student(Base):

__tablename__ = "students

" id = Column(String, primary_key=True, default=lambda: str(uuid.uuid4())) name = Column(String) Column(String) # stored as string class Attendance(Base): __tablename__ = "attendance" id = Column(String, primary_key=True, default=lambda: str(uuid.uuid4())) student_id = Column(String) timestamp = Column(DateTime, default=datetime.utcnow)

DATABASE CONNECTION (database.py)

from sqlalchemy import create_engine from sqlalchemy.orm import sessionmaker from models import Base

DATABASE_URL="sqlite:///./attendance.db"

engine= create_engine(DATABASE_URL) SessionLocal = sessionmaker(bind=engine) Base.metadata.create_all(bind=engine) def get_db():

db = SessionLocal()

Source Code Details

FACE PROCESSING (face_service.py)

import face_recognition import numpy as np def get_embedding(image_path):

image = face_recognition.load_image_file(image_path) encodings = face_recognition.face_encodings(image)

if len(encodings) == 0: return None return encodings[0] def compare_faces(known_embeddings, new_embedding, threshold=0.5):

distances = face_recognition.face_distance(known_embeddings, new_embedding) if len(distances) == 0: return None min_index = np.argmin(distances) if distances[min_index] < threshold:

return min_index return None

FASTAPI BACKEND (main.py)

from fastapi import FastAPI, UploadFile, File, Depends from sqlalchemy.orm import Session import shutil import numpy as np

from database import get_db from models import Student, Attendance from face_service import get_embedding, compare_faces app = FastAPI()

 # REGISTER STUDENT

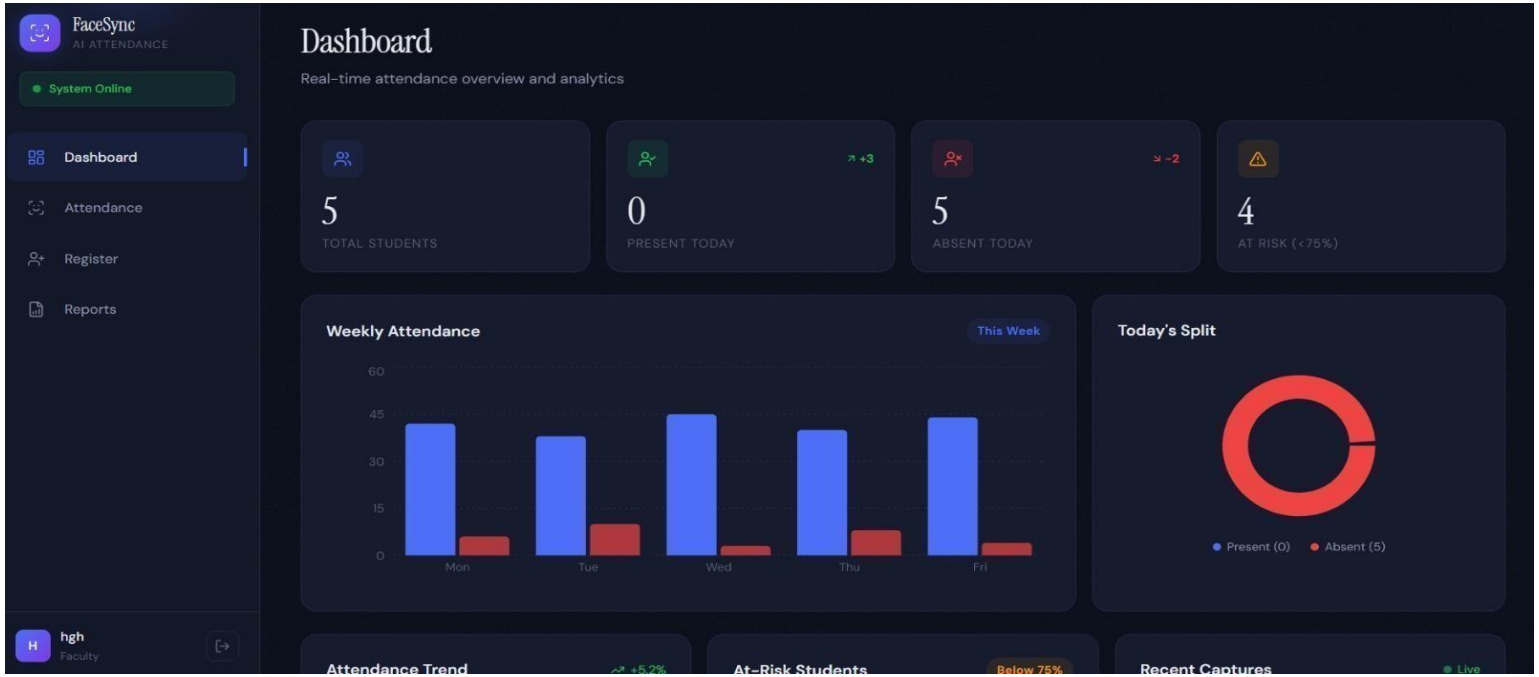
@app.post("/register") def register(name: str, file: UploadFile = File(...), db:Session = Depends(get_db)):

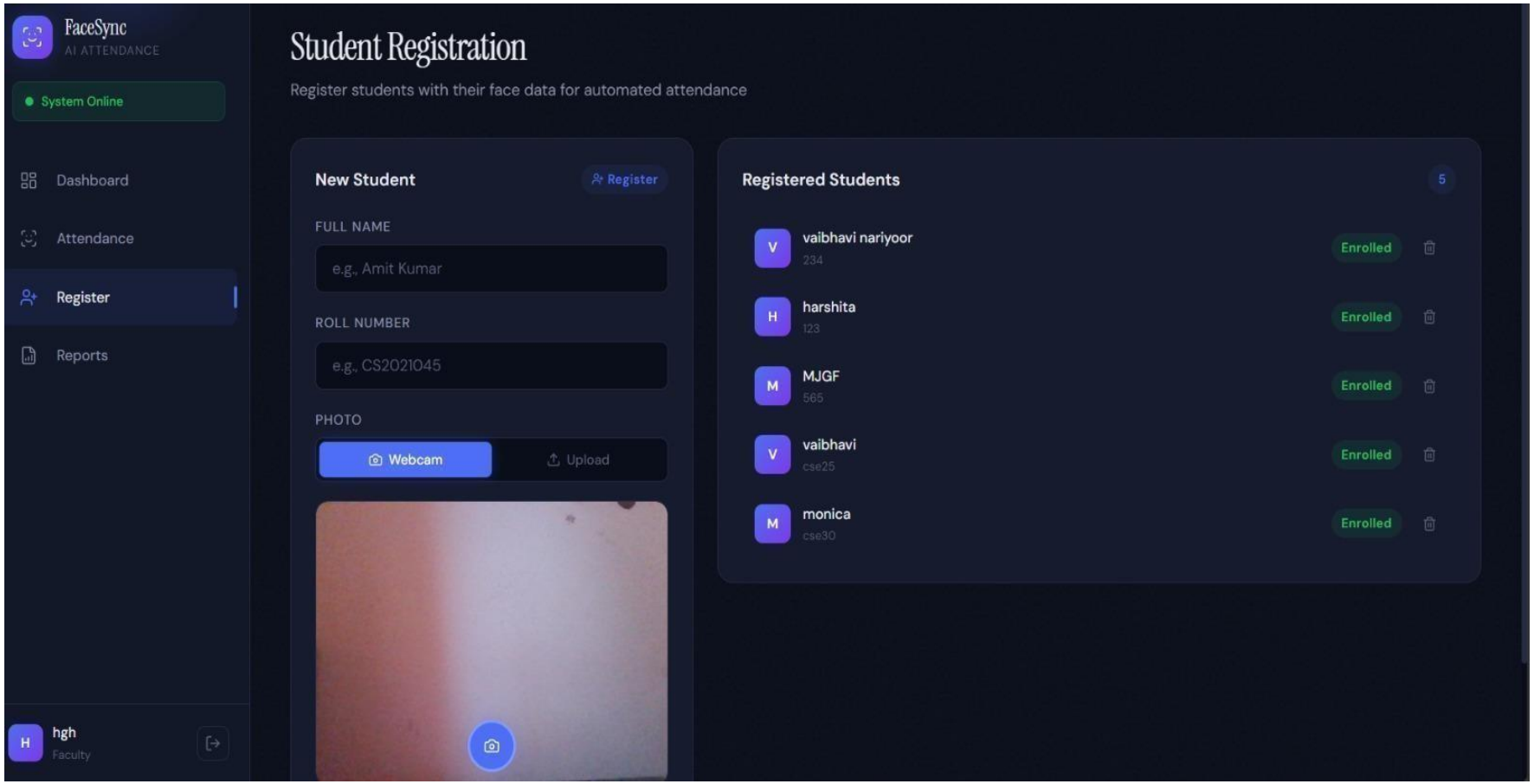
path = f"temp_{file.filename}" with open(path, "wb") as buffer: shutil.copyfileobj(file.file, buffer) embedding = get_embedding(path) if embedding is None:

return{"error": "No face detected"} student = Student (name=name, embedding=str(embedding.tolist())

db.add(student) db.commit() return {"message":

Results


FaceSync
AI ATTENDANCE
System Online
Dashboard
Attendance
Register
Reports
hgh
Faculty
Dashboard
Real-time attendance overview and analytics
5
TOTAL STUDENTS
0
PRESENT TODAY
+3
5
ABSENT TODAY
-2
4
AT RISK (<75%)
Weekly Attendance
This Week
60
45
30
15
0
Mon
Tue
Wed
Thu
Fri
Today's Split
Present (0)
Absent (5)
Attendance Trend
+5.2%
At-Risk Students
Below 75%
Recent Captures
Live


FaceSync
AI ATTENDANCE
System Online
Dashboard
Attendance
Register
Reports
hgh
Faculty
Student Registration
Register students with their face data for automated attendance
New Student
Register
FULL NAME
e.g., Amit Kumar
ROLL NUMBER
e.g., CS2021045
PHOTO
Webcam
Upload
Registered Students
5
vaibhavi nariyoor
234
Enrolled
harshita
123
Enrolled
MJGF
565
Enrolled
vaibhavi
cse25
Enrolled
monica
cse30
Enrolled

Github Link

The Github link provided should have public access permission.

Github Link : https://github.com/vaibhavi325

Thank
You!